Ex - 7

Sliding Window protocol

Aim:-

Write a program to implement flow control at data - Link Layer using sliding window protocol. Simulate the flow of frames from one node to another.

Sender program:

```
import time
w = int (input ("Window Size: "))
msg = input ("Message: ")
frames = [(i,c) for i,c in enumerate(msg)]
i=0
while i< len(frames):
    end = min (i+w, len(frames))
    with open("Sender_buffer.txt", "w") as f:
        for n,c in frames[i:end]:
            print ("Frame", n,":", c)
            f.write (f"{n}, {c}\n")
    time.sleep(2)
```

```
with open("receiver_buffer.txt") as f:
    ack = f.read().strip()
    if ack.startswith("ACK:"):
        a = int(ack[4:])
        if a == end:
            i = end
        else:
            print("Wrong ACK - resending")
    elif ack.startswith("NACK:"):
        i = int(ack[5:])
        print("NACK - Resending from", i)
print("Transmission complete")
```

receiver program:

```
import time
import os
expected = 0
while True:
    if not os.path.exists("sender_buffer.txt"):
        time.sleep(1)
        continue
```

Scanned with OKEN Scanner

```
with open("Send_buffer.txt") as f:
    frames = f.readlines()
if not frames:
    time.sleep(1)
    continue
error = False
for line in frames:
    n, data = line.strip().split(",")
    n = int(n)
    print("Received:", n, data)
    if n != expected:
        with open("Receiver_buffer.txt", "w") as f:
            f.write("NAck:" + str(expected))
        error = True
        break
    expected += 1
if not error:
    with open("Receiver_buffer.txt", "w") as f:
        f.write("Ack:" + str(expected))
open("Sender_buffer.txt", "w").close()
time.sleep(1)
```

Sender Input:-

Window size : 3

Message : HELLO

Sender output:-

Frame 0 : H

Frame 1 : E

Frame 2 : L

Frame 3 : L

Frame 4 : O

Receiver Input :-

sender_buffer.txt

0, H

1, E

2, L

3, L

4, O

Receiver output:

Received : 0, H

Received : 1, E

Received : 2, L

Received : 3 L

Received : 4 O

Result:

The sliding Window Protocol was successfully implemented to control frame flow & handle ACKs & retransmissions.

Scanned with OKEN Scanner